Industrial Safety Processes

Personal Protective Equipment (PPE)

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PPE Program

- Can be covered by one document or spread across multiple different documents and programs
  - PPE Safe Work Practice, Hearing Conservation Program, Respiratory Code of Practice, Glove Policy, etc.
- Needs to include PPE requirements, training, maintenance, inspections, disposal / repair
- Usually also incorporated in safety verification programs – Behaviour Based Safety, COR Audits, etc.

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Hierarchy of Controls

- Remember the hazard control hierarchy (elimination, engineering, administrative, PPE, combo)
- PPE is the last thing that should be considered and implemented as a hazard control measure
- However, can be used as a “temporary” measure pending the implementation of higher-order controls

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Identify PPE Needs in the Workplace

- Formal Hazard Assessments
- Manufacturer’s Operating Manuals (MOM)
- Industry Standards / Guidelines
- OHS Regulatory Requirements

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Considering Options

- Recommended to involve supervisors and front-line staff in considering options – sizes, colors, materials, etc.
- Can help with increased PPE use as required
- Note that some individual issues may need to be addressed (i.e., hearing aids, prescription glasses)

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Source and Acquire PPE

- Set specific purchasing standards for all types of PPE
- Permit some flexibility if your organizational culture and budget will allow
- Confirm PPE is adequate and meets both reference standards and organizational needs
- Consider an established PPE ordering and distribution program

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Training

- Staff need to be trained on the reasons for – and proper use – of all required workplace PPE
- Training needs to include:
  - Reasons for and limits of PPE
  - Proper use
  - End-user care, maintenance, and storage
  - End-user inspections
  - Repairs and removal from service

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Monitoring & Reinforcement

- Should be a regular element in formal hazard assessments, investigations, and incident reports
- Should include in internal system verifications – Behaviour Based Safety, site tours and observations, etc.
- PPE requirements – especially for high-hazard work – should be part of applicable toolbox talks or safety meetings

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Common Issues

- Who pays for what?
- What about different standards – ANSI, CSA, CEN, etc.?
- Recommended vs. required – inspections, certifications?

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Industrial Safety Processes

Personal Protective Equipment

Head Protection

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Types of Head Protection

- The hard hat (safety helmet) is the typical symbol of workplace safety
- Head protection comes in many forms, styles, and colours
- Like all PPE, should be selected for expected hazards, environment, and fit

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CSA / ANSI

- CSA / ANSI Standards for head protection are referenced in specific provincial and federal legislation
- Important to know what standards are in force in the jurisdiction in which you are working – not all the same
- Some items may be permitted in one province, but not another
- Clients / Prime Contractors / Constructors / etc. may also specify minimum or specific requirements (i.e., Type 2, Class E)

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Hard Hat Types & Classes

- Type 1 – overhead impact
- Type 2 – overhead and side impact
- Class G (General) – 2,200 volts
- Class E (Electrical) – 20,000 volts
- Class C (Conductive) - no electrical protection

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Hard Hat Colours

- Hard hat colouring does not signify anything unless specified by an organization
- Some regulations and activities specify colouring and reflective features for specific tasks (i.e., Traffic Control Person in BC)

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Hard Hat “Expiry”

- Hard hats generally do not have firm “expiry dates”
- Check the specific manufacturer’s operating manual (MOM) for your hard hat
- The date wheels or markings on a hard hat are manufacture dates – not expiry dates

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Inspection and Use

- Look for cracks, dents, damage
- Check shell and suspension – older suspensions can crack and split
- May need to remove surface dirt to get a better look
- Stickers may also need to be cleaned for visibility (reflective stripes) and / or identification (First Aider, Green Hand, Site Orientations)

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Sunlight

- Ultraviolet radiation and direct sunlight degrade hard hats over time
- Excessive heat will also damage the hard hat – especially if left inside vehicles in direct sunlight
- Discoloured hats should be removed from service

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Paint, Markings, Stickers

- The glue from stickers does not degrade hard hat shells
- But stickers may cover cracks or other defects
- Paint and markers may contain chemicals that do degrade hard hat shells, so best not paint or use Sharpies on hard hats

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Reversal

- Some hard hats – not all – may be worn backwards (brim to back)
- This may be necessary if workers are doing something that may cause the brim to get in the way (looking into survey equipment)
- Check to see if the hard hat allows reversal by design and is marked accordingly

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Care and Maintenance

- Usual procedure is to disassemble and wash with warm water and soap and let air dry
- Sweat bands may also be washed or replaced as needed
- Internal suspensions should also be replaced as they tend to become brittle over time

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Storage

- Hard hat should be stored out of direct sunlight and away from potential contaminants
- Hard hats should not be used to store safety glasses or gloves (or other potentially dirty items) when not in use
- Ideally hard hats should have a designated “place” for regular storage (kit bag, shelf) that protects it against damage

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Removal from Service

Hard hats should be removed from service if:

- They have stopped a object (falling / side strike)
- If the shell has cracks, deformities or is discoloured
- Whenever MOM says so
- Company has mandated a specific “expiry date” for hard hats (from date of manufacture)

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Lanyards & Chin Straps

- In some situations a hard hat lanyard or chin strap may be required
- Chin straps keep the hard hat on the head in situations where it may fall off – high winds, odd postures
- Lanyards keep hard hat from becoming a hazard to persons below the work area

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Bump Caps

- In some situations it is not practical to wear a full size hard hat (confined spaces, some industrial environments)
- Bump caps are a common solution to provide low-profile protection against accidental head strikes on objects
- Not sufficient for regular industrial / construction protection against fall objects or side impact – provincial regulations usually limit their application in a workplace

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Industrial Safety Processes

Personal Protective Equipment

Eye & Face Protection

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Standard PPE Element

- Common to industrial and construction sites
- Often inexpensive and easily replaceable
- Should be selected based on anticipated hazards, fit, and use
- Key issues – fogging, fit, permissible tinting of lenses, use of contacts or prescription eyeware

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Common Considerations – Eye Hazards

- Flying objects and debris
- Dusts, fumes, vapours
- Chemical contact
- UV radiation

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CSA / ANSI Standards

- Need to ensure that the safety eyewear meets or exceeds appropriate safety standards
- Typical requirements include impact resistance, side protection, and light filtration
- The colour of lens does not determine UV or heat protection – the chemical composition of the eyewear does

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Prescription Safety Glasses

- Prescription safety glasses as an option
- May need to balance cost against hazards and use
- Should meet or exceed same standards as regular safety eyewear

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Inspection, Use & Care

- Safety glasses should be free of defects, scratches, and clean
- May also be treated with anti-fog wipes if used in humid or wet environments
- They should fit well and not cause irritation, especially around the ears
- Special handling and care of prescription or high-end glasses (welding, laser) may be required

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Maintenance

- Most safety glasses are disposable and not meant to be fixed by end users
- High-end equipment (i.e., prescription glasses, advanced laser glasses) usually justify the cost of repairs by qualified personnel

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Face Shields

- Many varieties – construction and use based on work performed and expected hazards
- Simple face shields commonly require safety glasses to be worn under them
- Some face shield offer sufficient eye and face protection similar to safety glasses – see applicable references standards in your provincial OHS regulations

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Use, Care, Maintenance

- Same basic requirements as eye protection
- Higher-end equipment may required special handling & care – i.e., arc flash
- Often shields can be replaced if they become damaged or too degraded to see through properly

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Specialty Applications

- Welding and metal working
  - UV radiation, heat, molten metal, flying objects
- Forestry and wood chipping (flying objects)
- Chemical fabrication (splash)
- Medical (splash, droplet)

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Contact Lenses

- Lots of debate over the use of contact lenses with eye protection
- Major considerations are the nature of the hazard and potential for higher-than-normal negative outcome with contact lenses

NOTE: Canadian strict-liability for employers allows employers to more significantly restrict contact use in the workplace – workers cannot “assume the risk” and absolve the employer from responsibility in an incident.

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Industrial Safety Processes

Personal Protective Equipment

Hearing Protection

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Hearing Protection General

- Probably the most commonly misapplied PPE measure
- One of the reasons that occupational noise-induced hearing loss is increasing
- The implementation of a Hearing Conservation Program is an entire element

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Hearing Conservation Program

- Noise mapping or noise baseline established for workplace or work activities
- Work the hierarchy of controls
- Use hearing protection – if necessary – in combination with other controls (monitoring, training, work rotations, noise baffles, substituting equipment)

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Occupational Noise Levels

- Most provincial standards require hearing protection if noise is above 85dB(A) in an 8-hour shift
- Canada Labour Code Part 2 has a higher limit – 87 dB(A)
- Quebec – 90 dB(A)
- Higher noise levels – either episodic or throughout a shift – have lower time limit exposures and higher hearing protection requirements

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Noise Reduction Ratings (NRR)

- Commonly identified on approved hearing protection equipment
- To calculate the actual reduction in noise:
  - Take the NRR number
  - Subtract 7
  - Divide by 2

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NRR Calculation Example

Working with metal shearing machine over 8 hour shift.
Noise from the machine averages approximately 104 dB(A).

Use ear muffs with NRR of 30.

$$(30-7)/2 = 11.5$$

$$104 - 11.5 = 92.5 \text{ dB(A)}$$

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“Doubling Up”

- Doubling hearing protection (plugs & muffs) does not add a numeric equivalence of protection (20 + 20 = 40)
- You can only add 5 to the NRR rating of the highest rated equipment
- 23 NRR plugs with 25 NRR muffs equals NRR of 30
  - 25 + 5 = 30

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Types of Hearing Protection

- Enclosure
- Aural
- Superaural
- Circumaural

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Main Issues

- Used before any other controls considered
- Often higher-order controls are deemed unnecessary after hearing protection is in place
- Requires training and signage as required – most people don’t know how to wear plugs properly
- Often removed for communication – consider radio-compatible or cut-off ear muffs

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Main Issues

- Inserting and using correctly
- Cleanliness of reusable equipment
- Monitoring potential hearing loss – audiometric testing
- Monitoring workplaces for changes in noise levels
- May need to “double up” with plugs and muffs

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Personal Protective Equipment

Respiratory Protection

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General Respiratory Protection

- To be considered if higher-order interventions do not reduce risk to workers to acceptable levels (ventilation, etc.)
- Aims to reduce exposure to harmful substances or maintain supply of breathable, non-toxic air
- Can range from “dust masks” to SCBA / SABA units

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Code of Practice

- If respiratory protection is required in the workplace, for whatever reason, provincial OHS regulations require some form of Respiratory Protection Code of Practice
- This generally requires hazard assessments, workplace monitoring of contaminants, training, fit testing, and medical clearances / reviews

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Major Considerations

- Identifying potential hazards to workers
- Oxygen deficient or toxic / contaminants present
- Assigned Protection Factor (APF)
- Maximum Use Concentration (MUC)

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Assigned Protection Factor (APF)

"Assigned protection factors (APFs) are used to indicate the level of effectiveness a respirator provides to the wearer. An employer should ensure that the APF for an assigned respirator is adequate to provide protection from the identified hazardous contaminant."

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Hagan, Philip E., Montgomery, John F., & O'Reilly, James T., eds. (2015). *Accident Prevention Manual for Business & Industry: Engineering & Technology* (14th Edition). Itasca, Ill.: National Safety Council., p. 202.

Contaminants

- Particulate
- Fumes
- Vapours
- Gases

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Maximum Use Concentration (MUC)

"…the maximum atmospheric concentration of a hazardous substance from which an employee can be expected to be protected when wearing a respirator, and is determined by the assigned protection factor of the respirator or class of respirators and the exposure limit of the hazardous substance. The MUC usually can be determined mathematically by multiplying the assigned protection factor specified for a respirator by the permissible exposure limit (PEL), short-term exposure limit, ceiling limit, peak limit, or any other exposure limit used for the hazardous substance."

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Occupational Safety and Health Administration. (2009). *Assigned Protection Factors for the Revised Respiratory Protection Standard (OSHA 3352-02)*, p. 3.

Types of Respiratory Protection

- Air Purifying
  - Disposable, reusable, filtration system
  - Not to be used in oxygen deficient environments (< 19.5%)
- Air Supplying
  - Air supplied by local tank or hose to clean air source
  - For oxygen deficient or highly-toxic conditions

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Basic Respirators – Air Purifying

- Commonly referred to as “dust masks”
- General use for all sorts of applications
- Protection varies greatly – both in terms of particulate size and resistance to liquids (particularly oils)
- Some require fit testing – such as N95s
- Disposable after single use or when no longer in good working order

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Reusable Respirators – Air Purifying

- Contains facepiece and harness assembly which can be fitted with a range of cartridges
- Provides greater protection and flexibility across a range of environments and applications
- May include elements of face / eye protection as well

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Full-, Half- & Quarter-Face

- Full- and half-face respirators are common – quarter-face are rare
- The hazards and conditions will determine which style will be required
- May need spectacle kit for full-face respirators if contact lenses are not considered appropriate

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Powered Air Purifying Respirator (PAPR)

- In certain situations additional protection is required by using powered air purifying respirators (PAPRs)
- These usually contain larger portable air pumps with filters
- Useful in hot or cold environments when supplied air need to be heated or cooled
- Good option if great protection needed, but supplied air is not required

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Cartridges – Selection, Use, Care

- Cartridges should be selected for the particular contaminate and concentrations expected
- Many cartridges offer chemical (vapour / fume) protection as well as particulate filtration
- Always follow MOM
- Should be changed out regularly – do not rely on odour "breakthrough" to identify expired cartridges

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Numbering & Lettering

- Filtration of 95, 99 or 99.97% of a particular sized of particle (0.3 microns or larger)
- N – Not resistant to oil
- R – Resistant to oil
- P – oil-proof

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Numbering & Lettering Examples

- N95
  - Not resistant to oil, 95% efficiency for particulate 0.3µ or larger
- R95
  - Resistant to oil, 95% efficiency for particulate 0.3µ or larger
- P100
  - Oil-proof, 99.97% efficiency for particulate 0.3µ or larger

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Cartridges – Colouring

- The colouring scheme for cartridges is a manufacturer-specific decision – it is not regulated
- A purple cartridge from 3M (P100 + organic vapour / acid gas) may do the same thing another purple cartridge (only P100)

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Supplied Air

- To be used in Immediately Dangerous to Life and Health (IDLH) environments
- Where oxygen is deficient (< 19.5 %)
- Where toxic contaminants cannot be filtered out by air purifying respirators

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Positive Pressure and On-Demand

- Positive pressure systems maintain a constant level of air pressure inside the facepiece
  - Can be irritating to workers (SCBA / SABA)
- On-demand system (SCBA) only supply air when the worker breaths in (pulls or demands air)
  - May require additional training
  - Can freak out workers who think there is no available air

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Self Contained Breathing Apparatus (SCBA)

- The entire breathing air system is carried by the worker
- These systems provide the greatest amount of worker flexibility and range of motion
- Can be heavy and tiring to carry
- Limited air availability restricts time of activity

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Open vs. Closed Circuit SCBA

- Open circuit SCBA expels use air and does not recirculate in the system, using breathable compressed air
- Closed circuit SCBA uses a chemical source for oxygen and recirculates the air
- Closed circuit systems are extremely rare in Canadian industrial applications

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Supplied Air Breathing Apparatus (SABA)

- Breathable air is supplied to the worker from a stationary location, typically a bottle truck or compressor system
- Same application as the SCBA, but no heavy weights to carry and longer time available for work
- Does have limitation on worker mobility and activity – restricted by hose and conditions which may damage it

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Escape Systems

- SCBAs may have escape bottles included, but not common
  - There are fitted with alarms (audible and sometimes vibration) when the air supply is getting low
- SABAs must have escape bottles as part of the deployment in case the air supply is disrupted
- Typical escape bottles are rated for between 5 and 15 minutes of air

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Fit Testing

- Required for most forms of respiratory protection – including N95 disposable respirators
- Determines proper mask fit and effective seal
- Mask configurations vary between manufacturers, so fit tests are only good for specific makes and models

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Qualitative Fit Testing

- Typically used only for air purifying respirators
- Relies on the worker reporting smelling an irritating agent in a controlled environment (hood)
- Requires an irritant that the worker can identify in the absence of the respirator being tested

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Quantitative Fit Testing

- Specifically tests the air pressure inside and outside the respirator
- Checks fit and effective seal to ensure proper protection to the worker
- Does not rely on worker feedback
- If equipment properly working, gives accurate measurement of respirator effectiveness

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Repeating Fit Testing

- Normal fit testing is good for 2 years
- Changes to facial structure (weight gain/loss, loss of teeth, accident) may require an updated fit test
- Changes to overall health may require updated and more frequent testing (lung issues, diabetes)

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Pre-Use Checks

- Must be conducted every time respirator is put on
- Positive Pressure
  - Blow out to identify leaks from respirator seal
- Negative Pressure
  - Breath in and hold breath to see if air leaks inside

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Inspection, Care & Maintenance

- Disposable respirators should be visually inspected before use and disposed of if damaged or not working properly
- Reusable respirators of all kinds should be carefully inspected for any form of defect or problem
  - Follow MOM
- Reusable respirators should be cleaned often, usually after each use

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Inspection, Care & Maintenance

- Special care should be taken to ensure facepieces, harnesses, hoses, valves, and fitting are all in good condition
- Cartridges should also be inspected for damage
- Some parts may be replaceable by the end user (valves in air purifying respirators)
- Emergency response equipment should be inspected at least monthly

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Storage & Handling

- Follow MOM's recommendation for storage and handling of reusable respirators
- Be careful to avoid cross-contamination
- Cartridges may need to be removed and stored separately from respirators if they are meant to be reused
- Shared respiratory equipment needs to be cleaned after each use

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Industrial Safety Processes

Personal Protective Equipment

Hand Protection

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Gloves in General

- Hand injuries are very common in many industries
- Gloves need to be selected based on expected hazards
- Important to ensure gloves do not interfere with required work or create additional hazards
  - Kevlar, fine manipulation tasks, catching in rotating machines

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Glove Variety

- Gloves are available in a wide range of materials and styles
- Gloves can be resistant to cuts, abrasions, heat, vibration, impact, chemicals, electricity, biological hazards, etc.
- May need a range of glove types and styles to meet task requirements, regulations, client specifications, or worker preference

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Cut, Impact & Abrasion Resistance

- Gloves are available that meet strict standards for cut, impact and abrasion resistance
- Should consider these in purchasing standards – especially for specific tasks or groups of workers
  - Sharp object handling, heavy material handling, heavy industrial work

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Industrial Safety Processes

Personal Protective Equipment

Body & Limb Protection

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Arm Protection

- Typically used to protect against cutting instrument edges
- Also use as heat resistance, especially in industrial environments

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Torso Protection

- Often to keep workers clean and free of contaminants – dirt, chemicals, biological wastes
- Occasionally include elements of administrative / warning controls – reflective stripes
  - Can be the primary purpose, not a protective device per se
- Full body and may include additional impact protection

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Leg Protection

- Common in forestry operations, particularly chainsaw use
- May see in limited conditions in other industries where there is a risk of cuts to lower limbs
- Occasionally used by workers in demolition (jackhammers)

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Cold Weather

- Range of full body and specific body parts (legs, torso, head, hands)
- Need to consider other hazards that need to be addressed
  - Flame-resistance, flotation, impact
- Important that cold weather “additions” don’t compromise other protective measures (i.e., hard hat liners)

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Hot Weather

- Cooling Bandanas
- Sun shade attachments – hard hats
- Rain Gear

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Water Hazards

- Provincial regulations usually require personal flotation devices for workers working over water hazards
- Can be simple PFDs (personal flotation devices) or high-end survival suits
- May need to combine with other required PPE – fall protection

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Other Considerations

- Insect repellent is a form of PPE when it is applied to individuals
- Sunscreen is the same as insect repellant
- Barrier creams may also be used in limited situations for protection against specific chemical hazards
  - Not common and use is generally discouraged in favor of more substantial and reliable protection

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Industrial Safety Processes

Personal Protective Equipment

Foot Protection

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Foot Protection in General

- Safety boots are a staple at industrial and construction sites
- Vary based on a range of features – ankle height, types of protection, additional features (anti-slip)
- More than just “steel toes”

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CSA Symbols

- More than the green triangle

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Toe Caps (clown shoes)

- Temporary toe caps, often used for visitors to industrial locations
- Do not provide other forms of protection (i.e., metatarsal, sole punctures, static discharge)
- Not suitable for active construction sites

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Not always “Steel Toes”

- Different materials are used to make “steel toes” today
- Composite fibre, plastics, aluminum
- May need specific types of “toes” for different work
  - Composite toes and shanks for electrical work
- Many “composite” boots are composite only in the shank – need to confirm material for toe protection

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Inspection, Care & Maintenance

- Should be cleaned regularly
- Include removing caught items in sole treads
- Be cautious of employees continuing to wear boots with loose stitching – may cause failure under stress

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Removal from Service

- The “steel toes” need to be covered at all times – if you can see the toe cap through the leather, time to chuck
- Many boots now come with additional protection in the toe area to extend life
- Worn treads or punctured soles

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Traction Aids

- In Canadian winters traction aids are often valuable to prevent slips and falls
- Problem with movement from ice / snow to pavement, hard surfaces, steel walkways, and indoors
- Could represent an ignition source in some environments
- Many different styles and configurations – some required or banned from different operations

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